

Disaster Relief and Management System

1. Project Overview

The Disaster Relief and Management System is a Java-based desktop application connected to a MySQL database. It helps relief workers manage disaster victims, supplies, medical records, and inquiries during disaster situations. The system supports two types of workers — Central Workers and Location-Based Workers — each with different roles and access levels.

2. Technology Stack

Technology	Version	Purpose
Java JDK	21+	Core programming language
MySQL	8.0	Database management
Eclipse IDE	Latest	Development environment
MySQL JDBC	Connector/J	Java-MySQL connectivity
Java Swing / JavaFX	Built-in	Graphical User Interface

3. Project Structure

The project follows a layered architecture with the following package structure:

```
DisasterReliefSystem/  
src/  
com.disasterrelief.models/ – Java model classes (Person, Victim, Supply...)  
com.disasterrelief.dao/ – Database access layer (VictimDAO, SupplyDAO...)  
com.disasterrelief.interfaces/ – Java interfaces (ReliefWorker, CentralWorker...)  
com.disasterrelief.exceptions/ – Custom exception classes  
com.disasterrelief.utils/ – Utility classes (DBConnection)  
com.disasterrelief.gui/ – GUI screens (LoginFrame, VictimPanel...)  
sql/  
schema.sql – All 16 table definitions  
lib/  
mysql-connector.jar – MySQL JDBC driver
```

4. Database Schema

Database Name: DisasterReliefDB | Total Tables: 16

#	Table Name	Description
1	GOVT_AGENCY	Government agencies that declare disasters

2	DISASTER	Disaster events with type, severity, regions
3	PERSON	Base table for all persons
4	VICTIM	Disaster victims with injury and address info
5	INQUIRER	People searching for missing victims
6	SOCIAL_WORKER	Relief workers with specialisation
7	FAMILY_RELATION	Family relationships between victims
8	MEDICAL_RECORD	Medical treatment records for victims
9	VICTIM_DIETARY_RESTRICTIONS	Dietary restrictions (multivalued attribute)
10	LOCATION	Relief camp locations with capacity
11	RELIEF_SERVICE	Inquiry logs linking inquirer to victim
12	VENDOR	Supply vendors with ratings
13	SUPPLY	Supply items with quantity and expiry
14	LOCATION_SUPPLY	Junction: supplies stored at locations
15	VENDOR_SUPPLY	Junction: supplies provided by vendors
16	VICTIM_SUPPLY	Junction: supplies allocated to victims

5. Setup Instructions

Step 1: Install Java JDK 21+ from [oracle.com](https://www.oracle.com)

Step 2: Install MySQL 8.0 + MySQL Workbench from [mysql.com](https://www.mysql.com)

Step 3: Install Eclipse IDE for Java Developers from [eclipse.org](https://www.eclipse.org)

Step 4: Download `mysql-connector-java.jar` and add to Eclipse build path

Step 5: Open MySQL Workbench and run `sql/schema.sql` to create all 16 tables

Step 6: Open Eclipse and import the project

Step 7: Update `DBConnection.java` with your MySQL root password

Step 8: Run `Main.java` to launch the application